# Advanced Plane Geometry

**ANSWER KEY**

## Section 1: Circle Theorems & Power of a Point

**1** Two chords $AB$ and $CD$ intersect inside a circle at point $P$. If $AP = 4, PB = 9,$ and $CP = 3$, what is the length of $PD$?

**a.** 6

**b.** 8

**c.** 12 ✓

**d.** 15

By the intersecting chords theorem: $AP \times PB = CP \times PD \Rightarrow 4 \times 9 = 3 \times PD \Rightarrow 36 = 3PD \Rightarrow PD = 12$.

**2** From an external point $P$, tangent segment $PT$ touches the circle at $T$, and secant line $PAB$ passes through points $A$ and $B$. If $PT = 10$ and $PA = 5$, what is the length of $AB$?

**a.** 10

**b.** 15 ✓

**c.** 20

**d.** 25

By tangent-secant theorem: $PT^2 = PA \times PB \Rightarrow 10^2 = 5 \times PB \Rightarrow 100 = 5PB \Rightarrow PB = 20$. Therefore, $AB = PB - PA = 20 - 5 = 15$.

**3** Two secants $PAB$ and $PCD$ meet outside a circle. If the intercepted arcs are $\overset{\frown}{BD} = 120^\circ$ and $\overset{\frown}{AC} = 40^\circ$, what is the measure of angle $P$?

**a.** $30^\circ$

**b.** $40^\circ$ ✓

**c.** $50^\circ$

**d.** $80^\circ$

Exterior angle theorem: $\angle P = \frac{\overset{\frown}{BD} - \overset{\frown}{AC}}{2} = \frac{120^\circ - 40^\circ}{2} = \frac{80^\circ}{2} = 40^\circ$.

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## Section 2: Triangle Centers & Cevians

**4** In $\triangle ABC$, medians intersect at the centroid $G$. If median $AM$ has length $18$ cm, what is the length of $AG$?

**a.** $6$ cm

**b.** $9$ cm

**c.** $12$ **cm ✓**

**d.** $15$ cm

The centroid divides each median in a $2:1$ ratio from vertex to midpoint: $AG = \frac{2}{3}AM = \frac{2}{3}(18) = 12$ cm.

**5** Which triangle center is equidistant from all three sides of a triangle?

**a.** **Incenter ✓**

**b.** Circumcenter

**c.** Centroid

**d.** Orthocenter

The incenter is the intersection of the angle bisectors and is the center of the inscribed circle, equidistant from all three sides.

**6** In $\triangle ABC$, the circumcenter lies on the midpoint of the longest side. What type of triangle is $\triangle ABC$?

**a.** Acute

**b.** **Right ✓**

**c.** Obtuse

**d.** Equilateral

By Thales's theorem, the circumcenter of a triangle lies on a side if and only if the triangle is a right triangle, where the hypotenuse is the diameter of the circumcircle.

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## Section 3: Solid Geometry & Volume Formulas

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**7** A right circular cone has a base radius of $6$ cm and a slant height of $10$ cm. What is the total volume of the cone?

**a.** $96\pi$ **cm**$^3$ ✓

**b.** $120\pi$ cm$^3$

**c.** $288\pi$ cm$^3$

**d.** $72\pi$ cm$^3$

Altitude $h = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = 8$ cm. Volume $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(8) = 96\pi$ cm$^3$.

**8** A solid metal sphere of radius $3$ cm is melted down and recast into a cylinder of base radius $3$ cm. What is the height of the cylinder?

**a.** $3$ cm

**b.** $4$ **cm** ✓

**c.** $6$ cm

**d.** $8$ cm

Volume of sphere: $\frac{4}{3}\pi(3^3) = 36\pi$. Volume of cylinder: $\pi(3^2)h = 9\pi h$. Equating: $9\pi h = 36\pi \Rightarrow h = 4$ cm.

**9** If the linear dimensions of a rectangular prism are all multiplied by $3$, by what factor does its surface area increase?

**a.** $3$

**b.** $6$

**c.** $9$ ✓

**d.** $27$

Surface area scales with the square of the linear scale factor: $k^2 = 3^2 = 9$.

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## Section 4: Cyclic Quadrilaterals & Ptolemy's Theorem

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**10** In a cyclic quadrilateral $ABCD$, opposite angles are supplementary. If $\angle A = (4x + 10)^\circ$ and $\angle C = (3x + 30)^\circ$, what is the value of $x$?

**a.** 20 ✓

**b.** 25

**c.** 30

**d.** 35

$\angle A + \angle C = 180^\circ \Rightarrow (4x + 10) + (3x + 30) = 180 \Rightarrow 7x + 40 = 180 \Rightarrow 7x = 140 \Rightarrow x = 20.$

**11** In cyclic quadrilateral $ABCD$, sides are $AB = 2, BC = 6, CD = 7,$ and $DA = 3$. What is the product of the diagonals $AC \times BD$ by Ptolemy's Theorem?

**a.** 32 ✓

**b.** 35

**c.** 39

**d.** 44

Ptolemy's Theorem states: $AC \times BD = AB \times CD + BC \times DA = 2(7) + 6(3) = 14 + 18 = 32.$

**12** What is Brahmagupta's formula for the area of a cyclic quadrilateral with side lengths $a, b, c, d$ and semiperimeter $s$?

**a.** $\sqrt{(s-a)(s-b)(s-c)(s-d)}$ ✓

**b.** $\frac{1}{2}(ac + bd)$

**c.** $\sqrt{s(s-a)(s-b)(s-c)}$

**d.** $\frac{ab + cd}{2}$

Brahmagupta's formula for cyclic quadrilaterals is $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$.

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